Temporal Regime Detection Through Multidimensional Time-Cost Representation in Complex Systems

Rafael Felippe Voigt
Independent Researcher, Brazil
Contact: rafael@voigtcore.com.br

Abstract

This paper presents a scientific framework for detecting operational regimes in complex systems through the analysis of multidimensional temporal cost accumulation rather than event prediction. The proposed approach models system behavior as a continuous interaction between temporal yield and temporal debit, enabling early identification of coherence decay, regime saturation, and irreversible transitions. A four-dimensional temporal representation (Candle 4D) is introduced to formalize regime states and visualize system sustainability over time. Financial and socio-technical systems are used as illustrative domains, demonstrating that instability emerges as a delayed consequence of accumulated temporal inefficiency rather than stochastic shock. The framework shifts system analysis from predictive forecasting toward temporal governance and structural observability.

---

1. Introduction

Complex systems operating under continuous pressure—such as financial markets, digital infrastructures, and socio-technical networks—rarely fail due to isolated events. Instead, failure emerges from prolonged inefficiencies, hidden temporal costs, and gradual loss of internal coherence. Traditional analytical approaches emphasize prediction, volatility estimation, or probabilistic forecasting. While useful, these methods often fail to capture the structural degradation that precedes regime collapse.

This work proposes a shift in perspective: from predicting outcomes to observing how time is consumed, accumulated, and converted into systemic cost. By modeling time as an explicit operational variable, it becomes possible to detect regime transitions before visible failure manifests.

---

2. Temporal Cost as a Systemic Variable

We define **Temporal Yield (TY)** as the effective productive time generated by a system, and **Temporal Debit (TD)** as the accumulated inefficiency, delay, or

irreversibility embedded in its operation. System stability is governed not by instantaneous performance, but by the long-term balance between TY and TD.

When TD persistently exceeds TY, the system enters a regime of coherence decay. This process is cumulative and directional, meaning that recovery becomes increasingly expensive as temporal debt grows.

---

3. Regime-Based System Modeling

The framework defines four operational regimes:

- **Stable**: TY > TD, coherence preserved.
- **Gas**: High activity with rising TD, unstable equilibrium.
- **Torpor**: Minimal activity, suppressed dynamics, apparent calm masking structural degradation.
- **Nano**: Critical micro-regime characterized by irreversible transitions and third-order temporal effects.

These regimes are not discrete events but continuous states identifiable through temporal accumulation metrics.

---

4. Candle 4D Representation

To formalize regime observation, we introduce the **Candle 4D**, a multidimensional construct extending classical time-series representation. The Candle 4D integrates:

1. Temporal Yield (TY)
2. Temporal Debit (TD)
3. Drift (Δt imbalance)
4. Regime Dominance

This representation allows post-close extraction of regime composition, dominance percentage, and transition direction, enabling operational observability rather than speculative forecasting.

---

5. Application to Socio-Financial Systems

Financial markets are treated as socio-technical systems governed by time allocation rather than price alone. Volatility spikes are interpreted as delayed reactions to prolonged temporal inefficiency. The framework demonstrates that crises emerge not from randomness, but from accumulated temporal imbalance reaching structural limits.

The same logic applies to organizational systems, infrastructure networks, and automated decision engines.

---

6. Discussion

The proposed framework does not aim to predict specific events. Instead, it provides a mechanism to control the cost of time within complex systems. By making temporal inefficiency observable, decision-makers can intervene before irreversible transitions occur.

This approach aligns with emerging perspectives in systems governance, resilience engineering, and operational physics.

---

7. Conclusion

This paper introduces a temporal governance framework for complex systems based on multidimensional time-cost representation. By shifting focus from prediction to coherence observability, the model offers a robust alternative for understanding regime dynamics and systemic risk. The Candle 4D formalism provides a scalable, domain-agnostic method for detecting regime transitions driven by temporal accumulation rather than stochastic shock.

Keywords

Complex Systems; Temporal Cost; Regime Detection; Systemic Risk; Social Networks; Financial Systems; Time Governance

RAFAEL FELIPPE VOIGT:04395796940 Assinado de forma digital por RAFAEL FELIPPE VOIGT:04395796940
Dados: 2026.01.08 14:41:00 -03'00'